

Bioactive peptides: a review

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Abstract

Bioactive peptides are protein components which are inactive within the protein structure, and upon release by enzymatic hydrolysis, they exhibit special physiological functions. In the last years, the characteristics of bioactive peptides obtained from various plant, animal and microbial sources have received much attention. Bioactive peptides are produced using hydrolysis by enzymes extracted from plants or microorganisms, or digestive enzymes and fermentation by proteolytic starter cultures. The composition and sequence of the amino acids determines their different functions, including relaxing effects, solute binding properties, strengthening of the immune system, antioxidant, anti-microbial, anti-inflammatory, cholesterol-lowering and anti-hypertensive effects. Bioactive peptides are identified by different methods including membrane separation techniques and chromatography from protein hydrolysis products and using spectrometric techniques. The possibility of using bioactive peptides as health or therapeutic components depends on ensuring their bio stability, bioavailability and safety.

Introduction

Centuries of experience and research have shown that there is a direct connection between nutrition and health. The ability of foods and their compounds to enhance the overall quality of life has been the focus of scientists for years. Until now, many bioactive compounds have been derived from macro biomolecules (such as lipids and proteins). Although the compounds made from proteins are among the most important and commonly studied types, there are still many open questions. In the human body, many protein hydrolysis products have shown different health benefits. Protein hydrolysis products (bioactive peptides) are special protein components that have significant biological effects and exhibit a positive effect on the function or condition of the body, so they have health-giving effects and are economically very important due to their use in the production of super-beneficial foods and medicines (Shahidi and Zhong, 2008). The different physiological roles of peptides have made them a suitable choice for the manufacturing of therapeutic compounds (Akbarian et al., 2022). Different types of physiological activities of bioactive peptides have been reported, which depend on the type, number, sequence and properties of amino acids in the peptides (Danquah and Agyei, 2012).

From a nutritional standpoint, the bioavailability (=degree of biological availability) of peptides is higher than that of proteins and free amino acids. In addition, smaller peptides have fewer sensitizing effects compared to primary proteins, and for this reason, protein hydrolysis products are widely used in infant food formulations (Danquah and Agyei, 2012).

Concerning therapeutic applications of bioactive peptides, much attention has been paid to the use of bioactive peptides in the field of pharmaceuticals and the production of beneficial foods. Peptides are preferable to traditional drugs for the following reasons:

1. They have more specialized activity towards the target tissue and therefore have little or no toxicity effects and can be effective even at low concentrations (Akbarian et al., 2022). This feature is very effective in the treatment of chronic diseases.
2. Chemical compounds from synthesis that are typically used as medicine have a kind of

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DOI: 10.2478/ebtj-2023-0013

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cumulative effect in the organism. These chemical compounds may cause environmental problems due to their disposal while they are still active. On the contrary, bioactive peptides do not accumulate in organisms and are easily destroyed in the environment (De Castro and Sato, 2015). A number of positive physiological effects of bioactive peptides include antioxidant, antimicrobial, antihypertensive, immune system strengthening, etc. They were reported *in vitro* or *in vivo*, each of which occurs due to the effect on one of the body's systems (Figure 1). Due to the importance of natural antioxidants, many researches on the antioxidant activity of bioactive peptides have been carried out and natural antioxidants have been identified as suitable substitutes for artificial antioxidants. Also, many studies have been done about the antimicrobial properties of bioactive peptides. Bioactive peptides are compounds of natural origin that can be obtained from different sources (plants, animals, marine) was extracted and used in the formulation of food and medicine (Gobbetti et al., 2007; Bazinet and Firdaous, 2009). During the past two decades, special attention has been paid to bioactive peptides that are released from food proteins (Korhonen and Pihlanto, 2006). Bioactive peptides are inactive in the main sequence of proteins, but they show multiple properties after being released, which is achieved through enzymatic hydrolysis. Enzymatic hydrolysis is one of the methods used to improve Application and nutritional features of food proteins as well as the production of bioactive peptides (Marambe, 2011).

The results of studies by many researchers have shown that peptides obtained from different food sources have a wide range of multiple biological activities such as antioxidant activity, antihypertensive activity, antimicrobial and other characteristics (Akbari Adregani and Shabani, 2018).

Today, about 150 types of peptides are used in the pharmaceutical and medical industries (Lau and Dunn, 2018). So far, more than 60 types of drugs have been produced using bioactive peptides (Fosgerau and Ohshima, 2015). Bioactive peptides have valuable nutritional properties that can improve the health of different body organs. These compounds are especially used in the production of useful foods and many medicines (Shahidi and Zhong, 2008.) because small peptide particles cause less sensitivity compared to proteins (the main sensitivity is digestive sensitivity; in other words, peptides are more sensitive to proteins are easier to digest and absorb) so they can easily be used in infant food formulation (Agyei and Danquah, 2012). Peptides have a series of antimicrobial activities that can be obtained by the hydrolysis of food proteins and secondary metabolites of some bacteria, such as bacteriocins.

Lactoferrin is one of the most important antimicrobial peptides known, whose enzymatic digestion leads to the production of other antimicrobial peptides with greater killing and inhibitory activity compared to lactoferrin itself (Mirdamadi et al, 2017). Antimicrobial peptides prevent the growth of microbial cells and thus destroy many bacterial and fungal cells (Sanchez and Vazquez, 2017).

Antimicrobial peptides

Antimicrobial peptides are ancient (evolutionary) weapons (Moghaddam et al., 2023). Their widespread distribution throughout the world of animals and insects shows that antimicrobial peptides play a vital role in the evolution of multicellular organisms. Antimicrobial peptides are still effective defensive weapons, disproving the common belief that bacteria, fungi, and viruses can become resistant to any imaginable substance (Moghaddam et al., 2023). Antimicrobial peptides target the poorly understood "Achilles heel of the microbe," a key part of the microbe's cell membrane design that distinguishes microbes from multicellular animals and plants. The knowledge gained from extensive research has led to commercialization efforts to invent new disinfection treatments.

Peptides with antimicrobial activity can be obtained both through the hydrolysis of proteins in foodstuff and as secondary metabolites from bacteria (bacteriocins). Until now, peptides with different antimicrobial activity have been detected in milk, of which lactoferrin is one of the most important ones. Enzymatic digestion of lactoferrin leads to the production of other antimicrobial peptides with more killing or inhibitory activity than lactoferrin (Mirdamadi et al., 2017).

The mechanism of antimicrobial activity of these peptides was found to be diverse. Although most of them destroy the stability of the target membrane by creating a hole in the cytoplasmic membrane, a group of peptides by targeting intracellular molecules and disrupting the synthesis of protein, DNA, enzyme activity or cell wall lead to the destruction of the target cell (Nicolas, 2009).

By binding to the surface of the target bacteria, the pore-creating peptides disturb the activity of the target cell membrane and as a result lead to its destruction (Abdou et al., 2007) studies show that the amine terminal region in this group of peptides plays a role in the killing activity (Dionysius and Milne, 1997). Penetrating the cytoplasmic membrane, these peptides are usually short sequences that have the ability to pass through the target cell membrane and are unstructured in solutions and as soon as they contact the cytoplasmic membrane, they find a secondary structure of alpha helix or beta sheets. In this situation, a biphilic structure is seen in these peptides. In this way, the polar part with the polar head of lipids and the non-polar part with the acyl head of the membrane (Last et al., 2013). Study has shown that most of these peptides contain less than 92 amino acids and are rich in cationic and hydrophobic amino acids. Previous research has also shown that the short length of the peptide chain reduces the probability of the peptide binding to the membrane. Because for antimicrobial activity to occur, it is necessary for the peptide to fill the thickness of the cell membrane, which is about 42 Å. In addition, the holes created by small peptides will not be stable enough. On the other hand, reducing the length of the peptides reduces their tendency to create the second type of alphahelix structure. Research has shown that it plays an important role in the antimicrobial activity of peptides (Strömstedt et al., 2010). Peptides smaller than 3 kilo Daltons (kDa) obtained from the hydrolysis of camel milk

or from whey proteins by protease K have more antibacterial activity against *Escherichia coli* compared to other peptide components. The reason for that is the easier passage of small peptides through the membrane of bacteria compared to larger peptides (Salami et al., 2010). The peptide component (9-12 kDa) was obtained from the autolysis of the yeast *Cloromyces marcianus* at a concentration of 21.2 ± 3.13 mg/ml. It prevents the growth of both *Listeria monocytogenes* and *Staphylococcus aureus* bacteria, and at a concentration of 32.9 mg/ml, Bethratib has inhibitory activity of $2 \pm 82.44\%$ and $1.9 \pm 77.18\%$ against *Listeria monocytogenes* and *Staphylococcus aureus* (Mirzaei et al., 2015).

Investigations show that the inhibitory activity of bacteriocins, another group of antimicrobial peptides, depends on the composition of the culture medium and the growth conditions of the producing bacteria. NaCl concentration and greenhouse time are effective in the optimal production of pentocin bacteriocin. The two mentioned factors among other factors (amount of tween, amount of peptone, amount of yeast extract, size of inoculum) were selected as the most effective factors in the production of pentocin using the statistical program of the response surface method (Motahari et al., 2016; Mirdamadi et al., 2017).

Cancer is one of the main causes of death worldwide, and in the last decades, significant progress has been made in the field of its diagnosis and treatment (Ferlay et al., 2010; Nourbakhsh et al., 2018). Common types of cancer treatments include radiation therapy, chemotherapy, and their combination, which can lead to the creation of resistance, toxicity, and damage to healthy cells in addition to removing cancer cells (Harris et al., 2013). In general, today, finding a suitable, cheap and less-complicated treatment for various cancers is one of the main concerns of cancer research (Panahi et al., 2018).

One of these solutions is the use of anti-cancer peptides, which have attracted the attention of many scientists due to their selectivity towards cancer cells, fewer side effects, and their effectiveness (Hoskin and Ramamoorthy, 2008). One of the types of peptides used is the antimicrobial peptide, which is part of the innate immune response against microbes in many species. These peptides have a low molecular weight (10-40 amino acids) and an amphipathic and cationic structure that allows them to target the negative membranes of cancer cells (like bacteria). In recent decades, the use of these peptides as anticancer agents has been considered as a new treatment method (Wang and Wang, 2004; Ejtehadifar et al., 2017; Schweizer, 2009). Antimicrobial peptides can act against cancer cells through various mechanisms such as apoptosis, necrosis, cell membrane lysis, inhibition of angiogenesis, and activation of the immune system (Ejtehadifar et al., 2017; Schweizer, 2009; Chernysh et al., 2002).

Opioid peptides (Relaxing peptides)

Previous studies showed that a group of peptides can play an important role in the nervous system, and these were called opioid molecules (Teschemacher et al., 1997). Opioid peptides

are small molecules with two hormonal and neurotransmitter activities. Based on their biological activity, these peptides are divided into agonist and antagonist groups, which imitate the activity of opiates in the brain with their short amino acid sequences. Opioid peptides obtained from food usually have 4 to 8 amino acids. These peptides create different activities by connecting to dissimilar types of receptors. The control of bowel movement and sedative effect are among the activities of this group of peptides (Shrikant Sharma and Rana, 2011). Beta-casomorphins are the first known opioid peptides derived from food proteins from milk beta-casein protein. Beta-casomorphin-7 having the sequence Tyr-Pro-Phe-Pro-Gly-Pro-Ile is the most famous peptide of this family (Mirdamadi et al., 2017).

The sequence of this peptide may be slightly different due to differences in the structure of the casein protein in various breeds of cattle. Alphalactorphin "Tyr-Gly-Leu-Phe" and betalactorphin "Tyr-Leu-Leu-Phe" are opioid peptides derived from milk proteins (Chakrabarti and Wu, 2016). Enzymatic digestion of the proteins in wheat gluten yielded several opioid peptides whose amino acid sequence was determined. It differs from peptides derived from other sources (Rutherford-Markwick, 2012).

Opioid peptides that are formed in the body act as neuro-modulators, which alter the actions of other neurotransmitters in the central nervous system. As a result of this modulation, opioid peptides can—among other functions—induce pain relief and euphoria as well as affect certain behaviors, including alcohol consumption (Froehlich, 1997).

"Endogenous opioid peptides are produced and often released together with other neurotransmitter molecules in the brain, pituitary gland, and adrenal gland as well as by single neurons in the central and peripheral nervous systems. For example, neurons in the carotid body simultaneously release an enkephalin and dopamine" (Froehlich, 1997). Similarly, single neurons in the peripheral nervous system of rats simultaneously release an enkephalin and the neurotransmitter norepinephrine (Froehlich, 1997).

Peptides that modulate the immune system

There is a clear connection between human nutrition and immunity. Bioactive peptides derived from different protein sources have immune system enhancing effects in both *in vitro* and *in vivo* studies. Peptides that strengthen the immune system can be produced naturally during digestive digestion and thus affect immunological responses and cell function, or they can be produced during enzymatic hydrolysis in laboratory conditions and used as food and medicine. In addition to its usual nutritional roles, milk plays an important role in strengthening the immune system of babies at birth (Mirdamadi et al., 2017).

Previous studies show that bioactive peptides derived from casein or whey are effective in stimulation and proliferation of lymphocytes, macrophages, antibody synthesis, regulation of cytokines and phagocytotic activity. A group of these pep-

tides, such as beta casein f191, play a role in stimulating macrophages. Beta casein f63 is effective in stimulating phagocytosis. Although in vivo research shows that both mentioned peptides are ineffective in creating immunity in mice (Fiat et al., 1993). Also, studies found that peptides derived from milk proteins are also effective in the proliferation of human peripheral lymphocytes (Kayser and Meisel 1996), monocytes and Helper T cells are among the cells that interact with these peptides. Lactoferrin acts similarly to opsonin (Mohanty et al., 2016) after breaking down and creating β -lactoferrin by binding to neutrophils (Mohanty et al., 2016). Lymphocytes and the type of activity are affected by the concentration of the peptide. The types that have inhibitory activity at low concentrations and have stimulating effects at higher concentrations (Trompette et al., 2003). Angiotensin converting enzyme with cutting the carboxyl end turns angiotensin I into angiotensin II and as a result increases blood pressure. Antihypertensive peptides prevent blood pressure increase by inhibiting this enzyme. Most known antihypertensive peptides have been isolated from cow's milk casein. Although antihypertensive peptides such as alphalactorphin and betalactorphin are also better isolated from alphalactalbumin and lactaglobin proteins (Mohanty et al., 2016).

The relationship between the size of peptides and the inhibitory activity of angiotensin-converting enzyme peptides reported (Ruiz-ruiz et al., 2013). Also, research showed that peptide components were smaller than 3 and 9-3 kilodaltons when made through the hydrolysis of proteins of the yeast *Clauromyces marcianus* by trypsin and chymotrypsin enzymes. They have the highest ACE inhibitory activity (Mirzaei et al., 2016). Both regions of the mentioned enzyme (region C and N) have the active site containing His-Glu-XX-His sequence. These active sites are located in the gap between the two regions and are protected by the amine cap. This cap limits the access of large polypeptides to the active site and this issue can explain the greater effect of small peptides in preventing the activity of angiotensin-converting enzyme (Gobbetti et al., 2002).

Although detailed information about the effect of the structure on the function of this group of peptides is not available, research shows that binding of peptides to angiotensin-converting enzyme is dependent on the carboxyl end of the peptide, and peptides with hydrophobic amino acids in this section are stronger inhibitors. These amino acids actually interact with the active site of the enzyme. On the other hand, other research reports that most aromatic amino acids, including tryptophan, tyrosine, and phenylalanine, as well as proline, are effective on this interaction (Cheung et al., 1980). The ratio of hydrophilic/hydrophobic amino acids in the peptide sequence plays a key role in the level of ACE inhibitory activity. Because the hydrophilic amino acids limit the access of the peptide to the active site of the enzyme (Li et al., 2004).

Peptides that bind to minerals in proteins enter interaction with ions in these minerals through their side chain amino acids. For example, alpha casein and beta casein interact with divalent and trivalent cations such as Ca^{2+} . But in addition to pro-

teins, peptides also have mineral binding activity. For example, phosphopeptides derived from casein, which are called casein phosphopeptides, also have this activity (Walther and Sieber, 2011). These phosphopeptides play a role in maintaining calcium, phosphorus, and other mineral elements in a soluble form under intestinal pH conditions (Akbarian et al., 2023). This activity is a result of the presence of phosphorylated amino acid serine, which has the ability to create salt with minerals such as calcium. As a result of the enzymatic digestion of milk, a diverse group of these peptides are produced (Miquel et al., 2005). The anionic structure of these peptides protects them from further proteolytic digestion and leads to the creation of a soluble complex with calcium and, as a result, the formation of calcium phosphate, which is practically insoluble in water. The type of amino acid composition in the phosphorylated region plays an essential role in the level of calcium binding activity in this category of peptides (Gagnaire et al., 1996). These peptides have also been reported to be effective in preventing tooth decay, osteoporosis, insomnia, and elevated blood pressure (Akbarian et al., 2023). Studies in animals report the positive effect of these peptides on calcium absorption in animals (Akbarian et al., 2023). A group of researchers have shown that whey protein fermentation with *Lactobacillus heloticus* is effective in the proliferation of osteoblasts in vitro (Akbarian et al., 2022). There are other reports based on increasing the bioavailability of iron in laboratory mouse models (Korhonen and Pihlanto, 2000).

Antioxidant peptides

Peptides with antioxidant activity can be won from enzymatic hydrolysis of proteins contained in food (Daliri et al., 2021). Extensive research on antioxidant peptides derived from casein, whey proteins, soybean, rice bran, wheat protein, egg protein and marine products was performed (Chang et al., 2013; Zhang et al., 2013; Xu et al., 2015; Wattanasiritham et al., 2015; Cian et al., 2015; Ko et al., 2013; Nimalaratne et al., 2015). These peptides are effective with various mechanisms including the removal of free radicals, the inhibition of lipid peroxidation ((the attac on lipids containing carbon-carbon double bond(s), especially polyunsaturated fatty acids (PUFAs)), avoidance of chelation of metal ions and formation of additional compounds. In addition to being important in the survival of body cells, antioxidation is also important in food and in order to reduce the possibility of spoilage (Walther and Sieber, 2011). Most of the studied peptides in this group are derived from casein-as. Various studies have shown that peptides with low molecular weight have more antioxidant properties than types with high molecular weight (Daliri et al., 2021). This effect is related to the better reaction of active free radicals with small peptides and was previously reported about the peptide components obtained from the hydrolysis products of meat, barley and canola proteins (Bamdad et al., 2011; He et al., 2013; Alashi et al., 2014). Also, the results of the research showed that the antioxidant activity of the protein hydrolysis product of sardine fish waste increases with the increase in the degree of hydrolysis and suggested that short chain peptides are

better antioxidants compared to long chain peptides (Bougatef et al., 2010). Further studies showed that the smaller peptide component from 3 kilo Daltons obtained from *Saccharomyces cerevisiae* yeast protein hydrolysis by trypsin enzyme has more DPPH and ABTS radical inhibitory activity than other peptide components (Mirzaei et al., 2015). Hydrophobic amino acids increase the solubility of peptides in the lipid phase and their accessibility to hydrophobic radicals, thus increasing their antioxidant activity (Chen et al., 1996). The high reactivity of aliphatic groups in alanine and valine with hydrophobic polyunsaturated fatty acids has been reported (Qian et al., 2008). They have also reported the presence of the amino acid alanine in the terminal positions and proline in the sequence of peptides with antioxidant properties isolated from gluten (Suetsuna and Chen 2002). In addition, to the role of aromatic amino acids, especially tyrosine, in the antioxidant activity of peptides (Duan et al., 2014; Pena-Ramos et al., 2004).

Bioactive peptides were also shown to exhibit mineral-binding activity (Akbarian et al., 2023). Proteins basically interact with mineral ions through their side chain amino acids and form the so-called chelate (Akbarian et al., 2023). For example, alpha casein and beta casein proteins interact with inorganic cations such as calcium (Walter and Sieber, 2011). Peptides derived from casein are highly effective in maintaining calcium, phosphorus and other mineral elements in soluble form under intestinal pH conditions (Miquel et al, 2005; Table 1).

Peptone

Peptone is a compound with a limited number of amino acids, usually between 3 and 10 amino acids, which is obtained from the breakdown (hydrolysis) of protein, usually milk and animal meat (mostly of bovine and porcine origin). Other sources are also possible, e.g. bacterial single cell protein (SCP) (Lackner et al., 2022), yeast or plants. Microbial-derived peptone currently sees a strong increase and demand (Newsnet, 2023). Peptones act as nutritional mediators in the growth of bacteria and fungi. Bacteria and cells need a source of carbon and nitrogen for their growth and development, and peptones are the single most important and main source of nitrogen for their growth and development (Tuysuz et al., 2021). Peptones are usually prepared in the form of powder or granules by the spray-dryer method and are soluble in water and aqueous solvents.

Peptones are used in various industries such as vaccine and drug production institutes, cosmetics and hygiene, yeast and yeast production factories, production of bacterial and cell culture media, including in medical diagnosis laboratories, some food industries that produce processed foods and ready meals, as well as ice cream production and The production of supplements, concentrates, and animal and poultry feed and many other productions of biological (biological) products are widely used.

Peptones are among the most important organic nitrogen sources which are added to culture media for growing microorganisms. They are defined as protein hydrolysates that are readily soluble in water and are not precipitable by heat, by

alkalis, or by saturation with ammonium sulfate (Taskin and Kurbanoglu, 2011).

Peptones are proteins that have been broken down into smaller parts by enzymes or acids. In fact, the result of this decomposition is the production of peptides and amino acids. Tryptone is a special type of peptone that is actually a protein (usually the milk protein casein) that is digested by the enzyme trypsin. The composition of peptone depends on the protein source and how it is digested. These factors determine the penetration rate of amino acids and peptides. For example, a common type of peptone is the peptone obtained from the digestion of soy protein by papin enzyme. Compared to tryptone, this type of peptone has a lower weight percentage of total nitrogen, as well as a different distribution of amino acids and the amount of carbohydrates is more than its other differences. The proteose peptone fraction of milk is defined as the proteins/peptides soluble in 12% trichloroacetic acid (TCA) after milk has been heated to 95°C for 30 min and acidified to pH 4.6 (Deeth and Bansal, 2019). Heating followed by acidification precipitates caseins and the major whey proteins denatured by the heat treatment (O'Mahony & Fox, 2013). Proteose peptones account for 0.8-1.2 g/L of milk (Claeys et al. 2014; Sørensen & Petersen, 1993).

"Proteose peptones are a mixture of proteins/peptides. They can be classified as those resulting from proteolysis of caseins and those indigenous to milk. The major products of proteolysis in this fraction are proteose peptone 5 (PP5), proteose peptone 8 (slow) (PP8s), and proteose peptone 8 (fast)(PP8f). These are, respectively, the following fragments of β -casein resulting from plasmin-catalyzed proteolysis: f1-105/107, f29-105/107, and f1-28. Other minor fractions such as f29-105 and f29-107 have been identified; several other fractions are present in low concentrations" (Deeth and Bansal, 2019). As these peptides are casein proteolytic products they can be considered with the caseins (O'Mahony and Fox, 2013). The main proteose peptones in milk were found to be proteose peptone 3 (PP3) and osteopontin (Deeth and Bansal, 2019).

Production methods of bioactive peptides

As the most common method for producing bioactive peptides, enzymatic hydrolysis by digestive enzymes can be named (Marambe, 2011). Once the amino acid sequence of the molecules is known, peptides may be synthesized by chemical synthesis or recombinant DNA technology (Marambe, 2011). Chemical synthesis methods are suitable for the production of short sequences and recombinant DNA technology for the production of larger peptides (Moller et al., 2008).

Basically, bioactive peptides can be produced from precursor proteins in three ways (Figure 1): a- enzymatic hydrolysis by enzymes extracted from microorganisms or plants, b- enzymatic hydrolysis by digestive enzymes, c- fermentation by proteolytic starter cultures, in most studies, a combination (a) and (b) or (a) and (c) have been effective in the production of short chain peptides (Korhonen and Pihlanto, 2006).

Table 1. Some bioactive peptides isolated from different protein sources.

Protein source	Used microorganism or enzyme	Peptide sequence	Therapeutic use	Reference
Beta-casein-casein as1	<i>Lactobacillus GG</i> enzymes+pepsin & trypsin	YPFP	Opioid	Park and Nam (2015)
<i>Saccharomyces cerevisiae</i>	Trypsin	YGKPVAVPAR	antihypertensive and Antioxidants	Mirzaei et al., (2015)
Camel milk and cow milk	<i>L. rhamnosus</i> PTCC 1637	Peptides weighing 9-12 and Less than 9 kilo Daltons	Antioxidants and antihypertensive	Moslehishad et al. (2013)
Milk	<i>Streptococcus thermophiles</i> <i>Lactobacillus bulgaricus</i>	YPYY	antihypertensive	Tsai et al., (2008)
Sodium Caseinate	<i>L. acidophilus</i> subsp. ATCC 4356 lactis G	Peptides with a mo- lecular weight of less than 3 kilodaltons	immune system improvement	Stuknyte et al., (2011)
Hydrolysates of camel milk protein		KDLWDDFKGL and MPSKPPLL	Antidiabetic	Mudgil et al., (2018)
Hydrolysates of camel milk protein		LPVPG	Antidiabetic	Nongonierma et al., (2017)
Fermented goat milk	<i>Lactobacillus plantarum 69</i>	ND	ACE inhibitor	Chen et al., (2018)
Hydrolyzed goat milk		ND	Antimicrobial activity	Zanutto-Elgui et al., (2019)

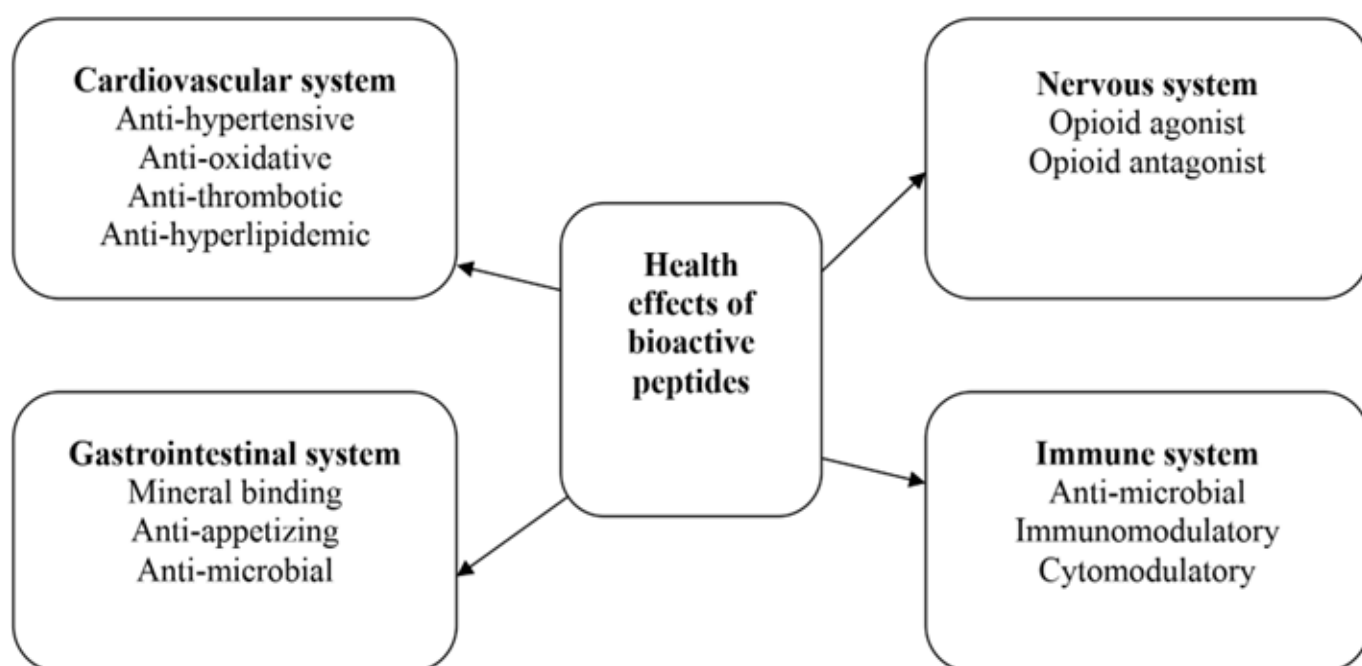


Figure 1. Physiological function of bioactive peptides

Enzymatic hydrolysis

Specific as well as non-specific proteases are preferably used to obtain bioactive peptides, due to a shorter time needed to reach the same degree of hydrolysis and better control over the hy-

drolysis process to get peptides with specific molecular weights and amino acid compositions. In this process, various enzymes such as pepsin, bromelain, trypsin, chymotrypsin and papain are used under their optimal pH and temperature conditions

(Singh et al., 2014).

Many of the known bioactive peptides have been produced, mainly using digestive enzymes, including pepsin and trypsin. For example, ACE-inhibiting peptides and calcium-binding phosphopeptides, CPPs) (CPPs are commonly produced by trypsin). Other digestive enzymes (Korhonen and Pihlanto, 2006) and different enzyme combinations of proteinases, including alcalase, chymotrypsin, pancreatin, pepsin, and thermolysin, as well as enzymes from bacterial and fungal sources, have also been used to produce bioactive peptides from different proteins (Kilara and Panyam, 2003). In addition, several enzymes with plant sources, such as papain and pronase have been used for enzymatic hydrolysis of soybean flour and wheat flour. Many bioactive peptides, such as biogenic, opioid, immunomodulatory, solute-binding, antihypertensive and antimicrobial peptides can be obtained by enzymatic hydrolysis of food such as milk, animal meat and fish, corn, wheat, soy and eggs (Singh et al., 2014).

In industrial applications, the use of stabilized enzymes is more common than conventional soluble enzymes due to several advantages (Akbarian et al., 2022). Immobilized enzymes can save costs to lower losses. They also enable enzymatic hydrolysis under milder and more controlled conditions than caustic or acidic hydrolysis. In addition, the stabilized enzyme can be recovered and the production of secondary metabolites caused by the autolysis of enzymes can be prevented (Agyei and Danquah, 2011). Also, industry typically relies on continuous hydrolysis processes, which is more effective and less expensive than conventional batch methods and is obtained by using reactors equipped with ultrafiltration membranes with different components, which may be combined with other purification techniques or units (Dionysius et al., 1997).

Fermentation by proteolytic starter cultures

Many dairy starter cultures that are used industrially have high proteolytic power. Therefore, bioactive peptides can be produced by both initiator and non-initiator bacteria used in fermented food products. Lactic acid bacteria, which are a large group of beneficial bacteria in nature and found in our digestive system, are used to produce bioactive peptides. Their role in the preparation of fermented products is not only because of their physiological effect but also because of their technological importance in the development of texture and taste (Singh et al., 2014).

The proteolytic system of lactic acid bacteria such as *Lactococcus lactis*, *Lactobacillus helveticus*, and *Lactobacillus delbrueckii* subspecies *bulgaricus* is now well known. This system consists of a protein attached to the cell wall and a number of intracellular proteins including endopeptidases, aminopeptidases, tripeptidases and dipeptidases (Christensen et al., 1999; Kunjiet et al., 1996).

Some studies have shown that the use of multiple fermentation as well as enzymatic hydrolysis in combination with the fermentation process increases the production of bioactive peptides. In a research, it has been reported that milk

fermentation with a commercial starter culture including five strains of lactic acid bacteria increases the ACE inhibitory activity (Chen et al., 2007). Treatment of milk with trypsin enzyme before fermentation with yogurt starter cultures leads to the creation of components rich in phosphopeptides. So, in the fermented samples treated with trypsin, the production of calcium phosphopeptides (CPP) β -CN (1-25)-4P and - α s1 CN (43-79)-7 was reported, while the amount of proteolysis was not significant in the samples that were fermented alone (Lorenzen and Meisel, 2005). High activity of inhibiting ABTS and DPPH radicals of camel milk fermented with *Leconostoc lactis* bacteria has been reported (Soleymanzadeh et al., 2016). In addition to living microorganisms, proteolytic enzymes isolated from lactic acid bacteria have been successfully used to create bioactive peptides from milk proteins. In a study, they measured the ACE inhibitory activity of casein hydrolysis product using 5 different commercial proteolytic enzymes. Among these enzymes, peptides produced from a protease isolated from *Aspergillus aureus* show the highest ACE inhibitory activity in vitro compared to other peptides (Mizuno et al., 2005). In comparison, microbial fermentation is a cheaper method to produce bioactive peptides than enzymatic hydrolysis. Because microorganisms are a cheap source of proteases and are known to be safe. The cost of bacterial cultures is relatively low due to the minimum nutritional requirements and short growth time, in addition, the proteases of lactic acid bacteria are expressed on the cell wall, which makes the extraction protocols relatively easy and cheap (Agyei and Danquah, 2011).

Concentration, purification and identification of bioactive peptides

The product of enzymatic hydrolysis of proteins contains a variety of peptides with different lengths and amino acid compositions. Therefore, separation and purification of bioactive peptides is an important part in the process of identifying bioactive peptides and determining their physical and chemical properties and evaluating their biological activity. Bioactive peptides are isolated from protein hydrolysis products by different methods, including membrane separation techniques and chromatography (Aluko, 2012).

In the membrane separation method, the protein hydrolysis product is poured on the surface of the porous membrane using pressure, and depending on the characteristics of the membrane, separation is performed based on the size of the peptide or electrical charge. Membranes are divided into types of microfiltration, ultrafiltration and nanofiltration. The most common types of ultrafiltration membranes are types with 1, 9, 3, 12 cut-off. However, other types of membranes, for example, with a cut-off of 2 and 1 kilodalton, have also been used in various researches to separate peptide components (Aluko, 2012). Reverse phase chromatography is the most common method for separating and purifying peptides, which allows for quick separation and identification of peptides and, in addition, determines their hydrophilic and hydrophobic properties. Chromatography device is usually used in combination with

quantitative and qualitative analysis equipment such as UV detector or mass spectrometry. Usually, a combination of several techniques is used to isolate and purify peptides. For example, first the peptide components are separated using membrane filtration and then purification is done using reverse phase chromatography. Protein sequencer and amino acid analyzer are the most common devices that to determine the composition of amino acids and their sequence. Other methods such as UV-Mass, fluorescence detection, Electrospray Ionization-tandem detection, and mass spectrometry are also used to identify peptides (Shahidi and Zhong, 2008).

Use of bioactive peptides in formulation functional foods

The presence of bioactive peptides as food components has been reported as health or therapeutic components in many fermented food products, and there is certainly a potential for the use of proteins and bioactive peptides as functional foods. People who are concerned about their health and people with special nutritional needs are the target groups for foods enriched with bioactive peptides. In addition, it is possible to add pure or synthetic bioactive peptides with different biological activities to different foods. For example, two peptides VPP and IPP with antihypertensive activity have been identified and commercially produced (Korhonen and Phlanto, 2006), but the use of bioactive peptides depends on solving problems such as biostability of peptides and their bioavailability.

When beneficial compounds are added to a food or beverage, they are subject to reactions with other compounds and may have significant effects on the overall quality of the product. Therefore, checking the stability and quality is the first step in the production of ultra-beneficial foods. The loss of one or a number of nutritional or flavor compounds or the creation of bad taste are factors that reduce the shelf life of products (Hamid et al., 2002; Abdul-Hamid et al., 2002).

The stability of bioactive peptides during the storage of food products is also an important issue that affects their bioavailability and factors such as light intensity, oxygen level, packaging permeability, temperature, relative humidity and internal factors including surface hydrophobicity, the presence of regenerating sugars, moisture content, pH and the degree of hydrolysis affect the stability of peptides (Rao et al., 2012; Day et al., 2009). For example, they investigated the effect of heat treatment in combination with acid and base on the inhibitory activity of ACE peptides derived from bovine casein. The results of their research showed that heating at a temperature of 122 °C at a pH of 5 to 12 decreased the activity of ACE inhibitory peptides and the appearance of the peptides darkened significantly. After heating in the presence of acid and alkali, the content of amino groups of peptides decreased significantly (Wu et al., 2014), but in the research of ACE inhibitory peptides obtained from dry-ripened Spanish meat, they have the same activity before and after heat treatment (from 92-117 degrees Celsius) in 12-3 minutes (Escudero et al., 2014).

For the biological activity of peptides, it is necessary for these compounds to remain in active form after digestive diges-

tion and be absorbed from the digestive system (Mirdamadi et al., 2017). Therefore, it is necessary to investigate the bioavailability of food bioactive peptides before their mass production. When bioactive proteins and peptides enter the digestive system after consumption, the function of digestive enzymes may affect their biological activity by changing their structure (Daniel and Kottra, 2004). The released peptides may show different functions. Some of them show their effects in the digestive system and some after absorption and distribution in the blood stream and reaching the target organ. Di and tripeptides can be absorbed intact from the digestive system (Shimizu and Son, 2007).

In addition, animal studies have shown that larger peptides (91-12 amino acids), resulting from the digestion of food proteins, can also be absorbed intact in the digestive tract and show their biological effects. However, the bioavailability of peptides and their absorption mechanism are still under study. The research showed that the activity of ACE inhibitory peptides obtained from Spanish dry-ripened meat did not change significantly after digestion by digestive enzymes in the in vitro environment (Escudero, Mora et al., 2014).

One of the practical methods to increase the stability of bioactive peptides is their microcoating. The microcoating of the compounds protects the sensitive components in the capsule and thus isolates them from the outside environment. This barrier provides protection against various factors such as oxygen, light and water and causes controlled release of compounds and prevents contact with other compounds. Another benefit of microcoating is the ability of controlled release of compounds, which provides their targeted transfer at the right time. The gradual release of compounds can increase the effect of food additives and ensure their dosage. The use of liposomes in coating proteins and other food components has been studied and different methods for the preparation of liposomes have been studied in different researches. Therefore, microcoating of protein hydrolysis products can protect them until they reach and release in food or human body (da Rosa Zavareze, Telles et al. 2014).

Safety of bioactive peptides

Although peptides have great potential to be used as food additives, they may not be suitable for all food applications, and it is important to check the safety and toxicity of these compounds (Phelan, Aherne et al. 2009). In general, there is little concern about the safety of bioactive peptides derived from food proteins. Because the body normally hydrolyzes food proteins into peptides and food grade enzymes and processes are used to produce bioactive peptides. Many studies have reported the non-toxicity of bioactive peptides in cell cultures (Udenigwe and Aluko, 2012). The issue of safety of peptides derived from milk has been investigated (Phelan, Aherne et al. 2009). In addition, new studies have shown that both single dose (2000 mg/kg) and repeated daily dose (1000 g/kg for four weeks) of casein hydrolysis products containing antihypertensive peptides have a negative effect on clinical parameters (including blood

biochemical properties, hematology, or histopathology) or mortality of mice. The amounts of consumed doses have been much higher than the amount that is necessary to perform pharmacological activities (Anadón, Martínez et al. 2010).

Conclusions

Bioactive peptides represent an interesting class of protein components that remain dormant within their parent protein structures until enzymatic hydrolysis triggers their physiological activities. The exploration of bioactive peptides sourced from diverse origins like plants, animals, and microorganisms (yeasts and bacteria), e.g. single cell protein, has gained significant traction. The potential integration of bioactive peptides as health-promoting or therapeutic agents hinges upon addressing concerns of biostability, bioavailability, and safety to ensure their suitability for such applications. As growth medium supplement, peptone is widely used.

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